



EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

GEOGRAPHY

6890/02

Paper 2 Geographical Skills

For examination from 2027

Total marks: 75

Specimen

MARK SCHEME

SECTION A

Answer **all** questions in this section.

- 1** The map extract is for Hwange West (Zimbabwe). The scale is 1:50 000.
Fig. 1 shows the position of some features in the central part of the map.

(a) Identify the following features shown in Fig. 1:

- | | | |
|---|-----------|-----|
| (i) name of river marked A ;
Chambumo | 1 @ 1mark | [1] |
| (ii) name of mountain B ;
Ndumebiza | 1 @ 1mark | [1] |
| (iii) mineral mined at C ;
coal | 1 @ 1mark | [1] |
| (iv) feature D ;
railway with embankment/embankment | 1 @ 1mark | [1] |
| (v) type of road E ;
narrow tarred road | 1 @ 1mark | [1] |
| (vi) type of road F ;
wide tarred road | 1 @ 1mark | [1] |
| (vii) feature G ;
gravel pit/quarry | 1 @ 1mark | [1] |
| (viii) the voltage of the power line at H ;
33kv | 1 @ 1mark | [1] |
| (ix) type of slope at I .
steep | 1 @ 1mark | [1] |

- (b) (i)** Give the six-figure grid reference of the railway bridge over the Deka river in the northern part of the map.

385731 1 @ 1mark [1]

- (ii)** Name **two** natural river features found along the Deka river.

- **river braiding**
 - **meanders**
 - **river confluence**
- 2 @ 1mark [2]

- (iii)** Give the type of drainage pattern in grid square 3370.

dendritic 1 @ 1mark [1]

- (vi)** Suggest the stage of the river Deka as shown on the map.

mature stage/middle course 1 @ 1mark [1]

- (c)(i) Measure the grid bearing of the spot height in grid square 3265 from the spot height in grid square 3567.

233° - 235° 1 @ 1mark [1]

- (ii) Measure the straight-line distance in **metres** between the two spot heights found in grid squares 3265 and 3567.

3590 - 3650 metres 1 @ 1mark [1]

- (iii) Calculate the gradient between the two spot heights in grid squares 3265 and 3567.

$$755-725 / 3600 = 30/3590-3650$$

=1:119-122 1 mark for correct calculation, 1 mark for correct answer [2]

Carry error forward from c(ii)

- (d) Identify **three** facilities found in Hwange town in grid square 4669.

- **post office**
- **police station**
- **district commissioner**
- **reservoir**

3 @ 1mark [3]

- (e) Using map evidence, list **two** advantages of Kamandama for the development of the mines.

- **availability of raw materials**
- **availability of transport /roads/railway**
- **availability of labour from town**

2 @ 1mark [2]

- (f) Study Fig. 2, which shows part of the map.

- (i) Identify the type of vegetation found in the shaded area.
medium bush

1 @ 1mark [1]

- (ii) Name relief feature **X**.
plateau

1 @ 1 mark [1]

[Total: 25 marks]

SECTION B

Answer **all** questions in this section.

2 (a) Study Fig. 3, which shows the climatic graph of a region.

(i) Name the climatic region shown in Fig. 3.

Equatorial forest region

1 @ 1mark [1]

(ii) Describe **two** features of the natural vegetation found in the climatic region shown in Fig. 3.

- **very tall**
- **bare**
- **straight**
- **some have climbers**

2 @ 1mark [2]

(iii) Use the information from Fig. 3 to complete Table 1.

Temperature (April) – 26°C

Rainfall (October) – 165mm

2 @ 1mark [2]

(b) Use Table 1 to calculate the following:

(i) The total annual rainfall received in this region. Show your working.

**185 + 175 + 165 + 190 + 150 + 155 + 135 + 125 + 145 + 165 + 170
+ 145 = 1905 mm**

1 mark for correct calculation, 1 mark for correct answer

carry error of reading graph forward

[2]

(ii) The annual temperature range. Show your working.

30°C - 25°C = 5°C

1 mark for correct calculation, 1 mark for correct answer

carry error of reading graph forward from (b) (i)

[2]

[Total: 9 marks]

3 Study Photographs **A** and **B** (Insert). Photograph **A** shows subsistence pastoral farming in an LEDC and Photograph **B** shows commercial pastoral farming in an LEDC.

(a) Give **two** examples of inputs for subsistence pastoral farming shown in Photograph **A**.

- **land**
- **grass**
- **water**

2 @ 1mark [2]

(b) List **two** characteristics of commercial pastoral farming like the one shown in Photograph **B**.

- **enclosed space**
- **feed provided**

- **special breeds**
- **farming products for sale/profit** 2 @ 1 mark [2]

(c) Using Photograph **A**, identify **two** problems faced by subsistence pastoral farmers.

- **overgrazing**
- **unfenced land**
- **bare land/soil erosion** [2]

(d) Suggest **two** solutions to problems faced by subsistence pastoral farmers.

- **rotational grazing**
 - **use of pesticides (tick control)**
 - **cattle breeding programs**
 - **fattening/Sisa ranches**
 - **education and training programmes**
 - **private industry assistance e.g. Eswatini Meat Industries.** 2 @ 1 mark [2]
- [Total: 8 marks]**

4 (a) Study Fig. 4, which shows an area with an informal settlement.

(i) What is an informal settlement?

an illegal/unplanned residential area usually found in urban area outskirts

1 @ 1mark [1]

(ii) Using Fig. 4, identify the main source of water supply in this settlement.

water standpipes

1 @ 1mark [1]

(iii) Using Fig. 4, name **two** features of the informal settlement which may lead to a rapid spread of diseases.

- **open drain**
- **marshy land**
- **near waste disposal site** 2 @ 1mark [2]

(iv) Suggest any **two** possible **negative** impacts of the informal settlement on the high-income residential area.

- **it is unsightly/visual pollution**
- **possible source of criminals**
- **lowers the property value of the high-income housing** 2 @ 1mark [2]

(b) Using Fig. 4, describe **two** ways by which this settlement could be improved.

- **construction of tarred roads**
 - **connect water supply in houses**
 - **increase number of grocery shops**
 - **more water standpipes** 2 @ 1mark [2]
- [Total: 8 marks]**

SECTION C

Answer **either** Question 5 **or** Question 6.

- 5** A group of students in the Island of Hawaii, an area affected by volcanic eruptions, decided to investigate the positive and negative impacts of volcanism on the people and the environment. They investigated the following hypotheses:

Hypothesis 1: *Volcanism is the main tourist attraction in Hawaii.*

Hypothesis 2: *Volcanism has a negative impact on the environment.*

(a) State **two** types of tourism.

- **coastal**
- **inland**
- **ecotourism**

2 @ 1 mark [2]

(b) Before carrying out the actual investigation, the students carried out a small pilot survey of the impact of volcanism on the people and the environment.

State **two** limitations of a pilot survey.

- **increases total costs of survey**
- **possibility of making inaccurate predictions of survey**
- **may take up time needed for actual survey**

2 @ 1 mark [2]

(c) To carry out the investigation, the students decided to use a map of Hawaii showing some tourist attraction areas (Fig. 5).

What type of data was the map used by the students?

secondary

1 @ 1 mark [1]

(d) To investigate **Hypothesis 1:** *Volcanism is the main tourist attraction in Hawaii*, the students designed a questionnaire, Fig. 6 (Insert).

(i) Using Fig. 6 (Insert), suggest **three** reasons why this is a good questionnaire.

- **brief/few questions (4)**
- **tick boxes/neat and proper layout**
- **questions are close-ended**
- **user friendly**

3 @ 1 mark [3]

(ii) Suggest why the students asked Question 1: "Are you a tourist or a resident in Hawaii"?

to find out if the interviewee is reliable/relevant for the investigation or not

1 @ 1 mark [1]

(iii) Results for Question 3: “Which of the tourist areas are you visiting?” are shown in Table 2. Use the data from Table 2 to complete the bar graph in Fig. 7.

1 mark each for correct plotting

1 mark for shading

[2]

(e) Write a conclusion to **Hypothesis 1: *Volcanism is the main tourist attraction in Hawaii.*** Use evidence from Table 2 and Fig. 7.

- **Hypotheses is correct/true**
- **most tourists come to visit the volcanic areas, 41% / 14 tourists out of 34**
- **fewer visitors to other tourist attraction centres e.g. waterfall 5, museum 4**

1 mark for taking a stand on the hypothesis

2 marks for evidence

[3]

(f) To investigate **Hypothesis 2: *Volcanism has a negative impact on the environment,*** the students interviewed 50 people and asked for their opinion on what they thought were the impacts of volcanoes. The students interviewed every 5th person they came across.

(i) What sampling technique did the students use?

systematic

1 @ 1 mark [1]

(ii) Give **two** advantages of the sampling technique mentioned in (f) (i).

- **eliminates human biasness**
- **ensures regular coverage of the study area**
- **saves time**

2 @ 1 mark [2]

(g) In conducting the interview, the following were identified by the residents of Hawaii to be the impacts of volcanism on the environment. The results of the investigation are shown in Table 3.

(i) Use the information from Table 3 to complete the pie graph Fig. 8.

1 mark for accurate sectors

1 mark for correct shading

[2]

(ii) Write a conclusion to **Hypothesis 2: *Volcanism has a negative impact on the environment.***

Use evidence from Table 3 and Fig. 8.

- **Hypotheses is true/correct**
- **percentage for negative impact on the environment are high/above 70% e.g. loss of natural vegetation $12/50 = 24\%$**
- **air pollution $8/50 = 16\%$**
- **1 mark for taking a stand on the hypothesis**

2 marks for evidence**[3]**

(iii) How can the investigation to **Hypothesis 2: *Volcanism has a negative impact on the environment*** be improved?

- **use other sources of data**
- **choose other sampling methods**
- **choose other volcanic areas**
- **investigation done by other students**
- **use secondary data**

3 @ 1 mark [3]

[Total: 25 marks]

6 Students in South Africa conducted research on international migration in their country. They investigated the following hypotheses:

Hypothesis 1: *Most international migrants in South Africa have migrated from neighbouring countries.*

Hypothesis 2: *Most international migrants in South Africa left their countries because of push factors.*

(a) (i) Define a *push factor*.

reasons/something/negative factors which make / encourage people to leave/ go away from an area / country.

[1]

(ii) State **one** problem that may be caused by immigrants to the receiving area.

- **overcrowding**
- **development of shanty towns**
- **crime**
- **competition for jobs**

1 @ 1 mark [1]

(b) The students visited a local town Piet Retief (Mkhondo) to investigate **Hypothesis 1: *Most international migrants in South Africa have migrated from neighbouring countries.*** A questionnaire was designed to collect the required data. They decided to interview 50 international migrants.

Fig. 9 (Insert), shows part of the questionnaire.

(i) List **two** disadvantages of using a questionnaire for this investigation.

- **migrants may be reluctant to answer the questions/ not talk to strangers/ may not cooperate**
- **migrants they may lie.**
- **language difficulties/ do not speak the local language/ English/cannot read or write/illiterate**

2 @ 1 mark [2]

(ii) The students agreed to use a random sampling method to select migrants to complete the questionnaire. Suggest how the students used this method of sampling.

- use random number tables to generate order in which to ask people/ pick numbers out of a hat
 - choose people who fit the sequence identified
 - ask anybody/ next person/ do not consider age or sex
- 2 @ 1 mark [2]

(iii) State **two** reasons why the students used the random sampling method.

- eliminates biasness
 - faster
 - simple
 - less knowledge required about respondents
- 2 @ 1 mark [2]

(iv) Suggest **one** other source of data students might have used.

- migration reports
 - population registers
- 1 @ 1 mark [1]

(c) For Question 1: "In which country were you born?", students used a map of Southern Africa, Fig. 10, as a reference.

The results for Question 1: "In which country were you born?", are shown in Table 4.

(i) Why were the migrants asked about their country of birth?

- to get reliable information/ to establish their country of origin or whether they are locals or not**
- 1 @ 1 mark [1]

(ii) Use the results in Table 4 to complete the block bar graph Fig. 11.

- 1 mark for correct plotting**
1 mark for correct key
- [2]

(iii) Write a conclusion to **Hypothesis 1: Most international migrants in South Africa have migrated from neighbouring countries.**

Use evidence from Fig. 10, Fig. 11 and Table 4.

- Hypotheses is correct or true
 - 40/50 or 80% of the migrants came from neighbouring countries
 - 10/50 or 20% of the migrants came from non-neighbouring countries
 - Fewer from distant countries such as Zambia, Angola and Malawi (compare neighbouring and non- neighbouring or distant countries)
- Allow comparison of individual countries e.g. Eswatini 20% while Zambia 8%**
1 mark for taking stand on the hypothesis
2 marks for evidence
- [3]

(d) The students further investigated **Hypothesis 2: Most international migrants in South Africa left their countries because of push factors** using the questionnaire in Fig. 9 (Insert).

(i) Suggest **two** advantages of the students collecting the questionnaire data themselves.

- **information could be up to date/more current**
- **students could speak directly to respondents to find out detailed reasons/ in-depth information**
- **reduces falsification (e.g., of demographic data)**
- **opportunity to modify/rephrase questions for benefit of respondent**

2 @ 1 mark [2]

(ii) Table 5 shows the results of Question 2 in the questionnaire: "What is the main reason why you migrated to South Africa?"

Use the information from Table 5 to complete the graph in Fig. 12.

2 marks for correct plotting of 'join family' data (width and length)

1 mark for shading

[2]

(iii) Write a conclusion to **Hypothesis 2: Most international migrants in South Africa left their countries because of push factors**. Use evidence from Table 5 and Fig. 12.

- **Hypotheses is true/correct**
- **more migrants came because of push factors 30/50 yet pull 20/50**
- **14 migrants left their countries because of unemployment (highest cause)**
- **9 left because of low paid jobs etc**

1 mark for taking stand on the hypothesis

2 marks for evidence

[3]

(iv) Suggest how the investigation to **Hypothesis 2: Most international migrants in South Africa left their countries because of push factors** can be improved.

- **use other sources of data**
- **choose other sampling methods**
- **choose other people**
- **investigation done by other students**

[3]

[Total: 25 marks]